

AirView AI

AI-Powered Urban Air Quality Intelligence for Smart City Intervention

Detailed document for ET AI Hackathon 2.0 - Problem Statement 5. AirView AI turns station data, weather forecasts, FIRMS thermal anomalies and geospatial source proxies into a professional smart-city command centre for forecasting, source-influence screening, enforcement prioritisation, intervention planning and multilingual citizen advisories.

Area	Final implementation
Mode	Live operational path using Open-Meteo AQ/weather + OpenAQ when available, with validated historical replay fallback
Cities	Delhi NCR, Agra, Amritsar, Lucknow, Ludhiana
Forecasts	PM2.5 and PM10, 24/48/72 hour horizons, current/future forecast path, actual-vs-predicted replay validation
Geospatial intelligence	1 km intervention grid, station marker, bounded FIRMS evidence, OSM evidence where available, no fake plume
Decision support	Source influence scores, enforcement priorities, interventions, scenario sensitivity, English/Hindi/Punjabi advisories

Important usage note

The demo should be presented as an operational prototype with live-provider support and a replay fallback. When live providers are unreachable, the interface clearly switches to historical replay. The product does not claim regulatory source apportionment, causal intervention impact, live population exposure, or Sentinel/GHSL coverage where the data is unavailable.

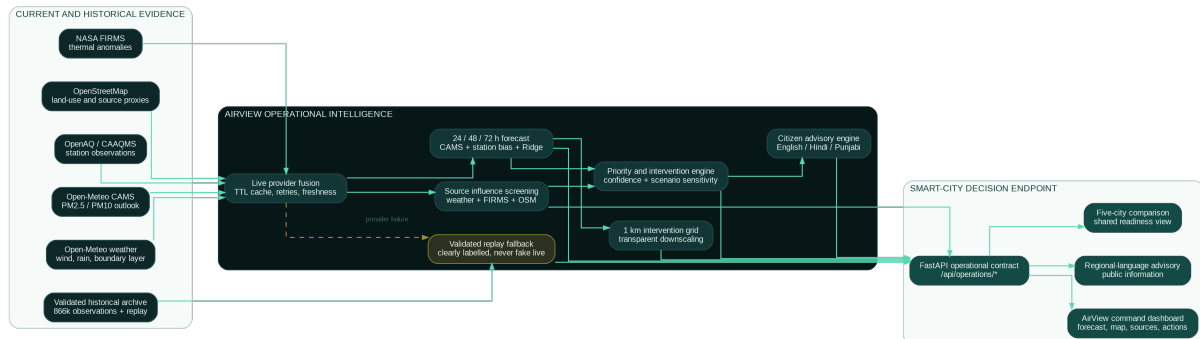
1. Problem fit and value proposition

The PS5 challenge asks for an intelligence layer that moves city administrations from reactive monitoring to proactive, evidence-based intervention. AirView AI addresses this by integrating monitoring observations, forecast providers, weather, thermal anomalies and geospatial source evidence into one operational workflow.

PS5 requirement	AirView AI implementation
Real-time IoT / CAAQMS integration	OpenAQ station integrations and historical CAAQMS-derived archive; OpenAQ recent/latest access used when live key/network are available.
Predictive analytics	24/48/72 hour PM2.5 and PM10 forecasts; fuses provider AQ forecast, observed station bias and trained local/persistence models.
Geospatial intelligence	Leaflet/OpenStreetMap map, station-centred 1 km planning grid, OSM source proxies, FIRMS event markers.
Atmospheric dispersion modelling	Practical wind, stagnation, humidity/rain and distance-decay screening for intervention prioritisation; not a regulatory dispersion model.
Multi-agent AI systems	Structured agents for readiness, forecast, intelligence, enforcement, intervention, advisory and audit trace.
Citizen communication	Template-based English/Hindi/Punjabi advisories with public-health disclaimers and audience variants.

2. System architecture

The production-facing architecture keeps provider ingestion, model evaluation, operational fusion, geospatial evidence, intervention logic and frontend presentation as independent layers. This makes the current five-city implementation scalable to additional cities without frontend redesign.

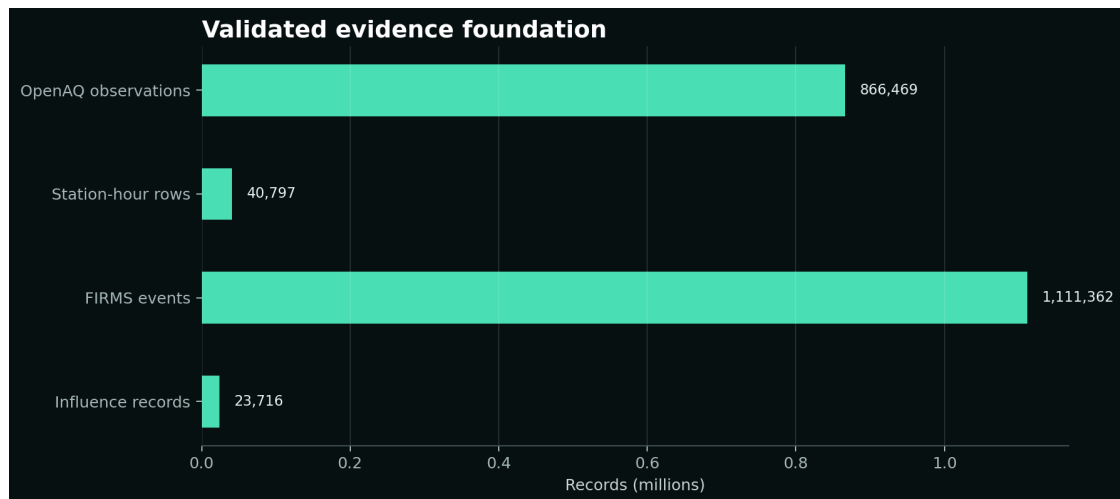


Architecture notes

The backend exposes a compact `/api/operations/dashboard` contract. It avoids sending entire historical Parquet files, FIRMS archives or development logs to the browser. The frontend only shows demo-ready decision views and excludes data-readiness/debug pages from the main presentation flow.

3. Data and evidence foundation

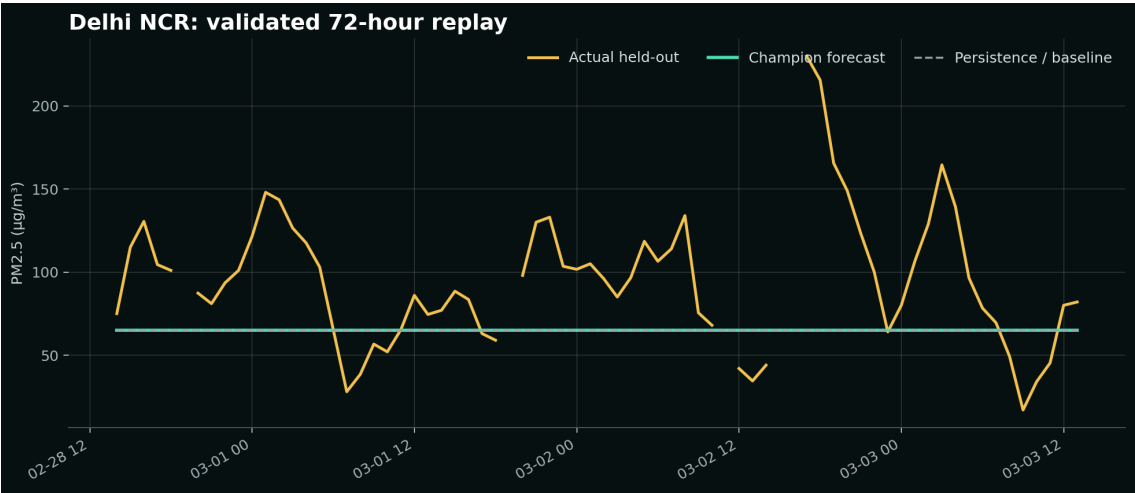
Source	Status in final prototype	Purpose
OpenAQ / CAAQMS-derived archive	866,469 historical observations, 40,797 station-hour rows	Replay validation, model features and current station integration path
Open-Meteo AQ and weather	Live no-key provider path in code	Current/future PM and weather forecast baseline
NASA FIRMS	1,111,362 canonical events, bounded map markers	Thermal anomaly and regional burning influence context
OpenStreetMap	Available for Delhi NCR, Lucknow and Ludhiana; unavailable for Agra and Amritsar	Traffic, industrial, construction and vulnerable-location source proxies
Sentinel-5P	Authenticated tests returned zero valid pixels; closed unavailable	Documented limitation, not fabricated
GHSL	No official raster present	Population exposure deferred until official raster is supplied



4. Forecasting and live-operation path

AirView AI now supports two paths. In live mode it attempts OpenAQ recent observations, Open-Meteo AQ forecast and Open-Meteo weather forecast. It fuses provider forecast, observed station bias and trained local/persistence corrections into 24/48/72 hour trajectories. If live providers fail, the validated historical replay loads automatically with clear labels.

Feature	Implementation
Current reading	OpenAQ recent/latest if fresh; otherwise Open-Meteo modelled current; historical replay fallback if providers fail
Future forecast	72-96 hour provider trajectory adjusted with observed bias and stored champion/persistence model signals
AQI-style severity	CPCB-style PM2.5/PM10 sub-index; clearly labelled exploratory when hourly values are used
Uncertainty	Operational interval from provider/model spread and replay residuals; shown as a band in UI
Validation	Replay comparisons include actual held-out values, persistence baseline and model prediction



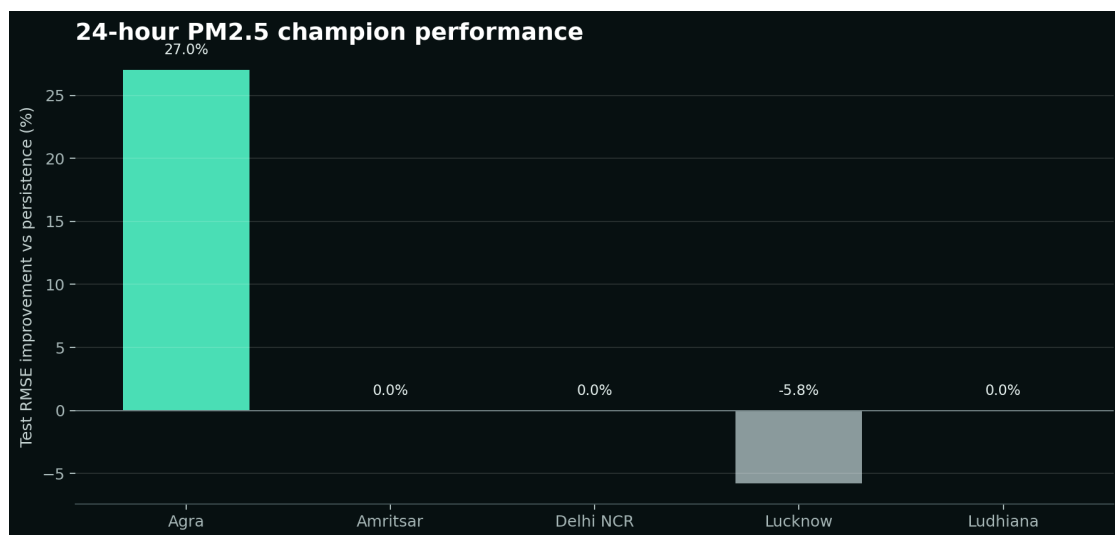
5. Command centre workflow

Step	Output
Select city/pollutant/window	Validated five-city controls: Delhi NCR, Agra, Amritsar, Lucknow, Ludhiana; PM2.5/PM10; 24/48/72h
Forecast panel	Actual/replay, AirView forecast, provider/persistence baseline and uncertainty interval
Map panel	Station, 1km intervention grid, FIRMS markers, OSM evidence where available
Source influence	Ranked influence indicators with confidence and missing-evidence disclosure
Action queue	Inspection priority, controllability, agency type, caveats and scenario sensitivity
Citizen advisory	Short public guidance in English/Hindi/Punjabi with replay/live status and disclaimer

The frontend is intentionally reduced to a single professional demo surface. Development logs, raw validation reports and fragmented prototype pages are not part of the user-facing workflow.

6. Evaluation and safeguards

Check	Final state
Backend/ML tests	26 tests passing in the delivered environment
Frontend quality	ESLint, TypeScript and production build pass
Operational endpoints	/api/operations/status, /cities, /dashboard and /network validated in demo mode
Map safety	FIRMS markers are bounded; no full FIRMS dump; no fake plume/heatmap
Ludhiana correction	Live Ludhiana uses PAU station; old Jalandhar replay is clearly disclosed and not mixed with live OSM evidence
Safety wording	Source influence is screening, not source apportionment; scenarios are sensitivity estimates, not causal effects



7. Demo and deployment

Run the demo locally from the repository root:

Command	Purpose
scripts/run_demo.ps1	Starts FastAPI backend and Vite frontend using the current source and saved artifacts
scripts/verify_demo.ps1	Checks health, operations API, five demo dashboards and frontend routes
scripts/stop_demo.ps1	Stops only demo-owned backend/frontend processes

Recommended demo path

Open /dashboard, choose Delhi NCR or Lucknow, select PM2.5, 72 hours, and Live first. If online feeds are not available on the demo machine, switch to Replay mode and explain that replay demonstrates the full verified pipeline with actual held-out observations.

Remaining limitations

Detailed outputs currently cover five validated cities. Operational quality depends on provider availability. Agra and Amritsar have no OSM spatial evidence in the final data package. GHSL population exposure and Sentinel-5P valid-pixel evidence are unavailable. The prototype is a decision-support system, not a regulatory enforcement authority.